# Computer Vision Group Project

ARI3129

Assignment Report

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# Contents

# 1 Introduction

The Aim of this project was to implement and evaluate various object detection algorithms on a custom dataset. The project was split up into the following sections:

- Building a dataset:
  1. Image collection,
  2. Image cleaning,
  3. Image annotation,
  4. Data analysis, and
  5. Data preprocessing.
- Object detection/ localization:
  1. Implementation of two object detectors to detect sign type,
  2. Implementation of one object detector to detect a specific sign attribute, and
  3. Evaluation of the trained detectors, including metrics and visualizations.
  4. Analysis of input image, supporting the maintenance and monitoring of traffic signs.

The following steps were taken to ensure an equal and fair distribution of work among all group members:

- For task 1, each group member contributed a similar number of annotated images to the dataset.
- For task 2, each team member was assigned one YOLO-based detector and one COCO-based detector to implement and evaluate.

# 2 Background on the techniques used

# 3 Data Preparation

To prepare the dataset for training, the following steps were taken:

1. The images were collected manually from diverse locations on the island, and at different times of the day. The goal was to capture a solid variety of sing conditions, angles, and lighting scenarios.
2. The collected images were then cleaned to remove any duplicates, after which, any recognizable faces or licence plates were obscured to ensure privacy. This was done both manually and using automated tools that can be found in the 'Annotation Process' folder of the GitHub repository [1] (all automated outputs were manually verified before moving onto the next step).
3. The images were then annotated manually using LabelStudio as instructed.
4. todo: about merging datasets
5. The combined dataset was then zipped and uploaded to the shared repository folder for easy access by all group members.

# 4 Implementation of the object detectors

## 4.1 YOLO-based Detectors

### 4.1.1 YOLOv8

### 4.1.2 YOLOv11

### 4.1.3 YOLOv12

## 4.2 COCO-based Detector

### 4.2.1 EfficientDet

### 4.2.2 Faster R-CNN

### 4.2.3 RetinaNet

### 4.2.4 RF-DETR

# 5 Evaluation of the object detectors

# 6 References and List of Resources Used

## References


[1] A. Spiteri, D. Pace, E. R. Debrincat, and J. Grech, “Annotation Process Tools,” GitHub repository, spiteriamy/ari3129-cv-group-project, Jan. 2026. [Online]. Available: https://github.com/spiteriamy/ari3129-cv-group-project/tree/main/Annotation%20Process